HEART RATE VARIABILITY AND PAIN: A SYSTEMATIC REVIEW AND META-ANALYSIS PROTOCOL

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Abstract

Background: Establishing a reliable pain biomarker may significantly improve care for nonverbal patients otherwise unable to express their pain (e.g., individuals with profound autism, cerebral palsy, aphasia due to stroke, or profound intellectual disability). Amidst potential candidates, heart rate variability (HRV) has garnered growing attention. However, our current understanding of the HRV-pain nexus remains rudimentary, which is limiting the integration of HRV as a standard pain measure in nonverbal patients within clinical settings. In this systematic review and meta-analysis, we aim to elucidate the association between HRV and pain.

Objective: The objective of this study is to assess the utility of HRV as an indicator of acute and long-term pain. To adequately address this objective, we have formulated two questions: 1) What is the relationship between HRV and experimentally induced pain? 2) How do chronic pain patients compare to healthy individuals with respect to HRV?

Methods: This meta-analysis will incorporate studies employing a range of HRV and pain intensity metrics, including both vagally-mediated and baroreflex-mediated HRV measures. We will employ recently developed techniques for detecting and adjusting for publication bias. Additionally, we will perform moderator analyses to identify potential heterogeneity sources among the selected studies.

Discussion: The outcomes of this meta-analysis may enhance our understanding of the relationship between pain and autonomic cardiac control, as measured by HRV. The findings may also aid the development of sophisticated tools for detecting, managing, and treating pain in patients who are unable to verbally communicate their pain.

Keywords: Pain, Heart Rate, Heart Rate Variability, Meta-Analysis, Systematic Review, Chronic Pain, Pain Biomarkers

Background

Identifying a reliable biomarker of acute and prolonged physical pain may markedly improve the care given to nonverbal people[1]. Individuals with profound autism*, intellectual disability, cerebral palsy, or aphasia due to stroke are at risk of being unable to express that they are in pain[2–6]. As a result, caregivers in facilities for non-verbal patients grapple with the challenge of discerning whether their patients are in pain[7–13]. Functional interactions between the cardiovascular and pain regulatory systems appear to be a crucial part of the pain regulatory process[14], and measures of this process may prove to be a dynamic biomarker of pain. Heart rate variability (HRV) is a measure of beat-to-beat variation in the heart rhythm over time. This reflects parasympathetic nervous system activity[15,16], and is an indirect measure of baroreflex function[17]. Baroreflex function refers to the reflexive decrease in breathing rate in response to sudden changes in blood pressure[18]. In recent years, HRV has emerged as a potential biomarker for both acute and chronic pain[19–21], with reduced HRV linked to chronic pain[19,22–24]. Thus, HRV might have the potential to aid patients who cannot otherwise communicate that they are in pain. Yet, HRV is not widely used as a measure of pain in patients who are unable to communicate their pain verbally[25]. Due to its complexity, our current understanding of the relationship between HRV and pain remains limited[19,21,26,27].

HRV can be assessed in time and frequency domains. In the time domain, interbeat interval data collected during resting conditions reflect parasympathetic cardiovagal outflow; and in the frequency domain, power spectral analysis of low frequency fluctuations provides a measure of baroreflex function[17]. In studies on HRV and pain, typically reported measures include:

* We recognize the difference in preferences for terminology within the community, but we have chosen the person-first language due to recommendations from our user representatives' organization, The Norwegian Autism Society (Autismeforeningen).

1. Vagally-mediated HRV measures (respiratory sinus arrhythmia (RSA); the root mean square of successive differences between heartbeats (RMSSD); and the high-frequency component of the power frequency spectrum (HF power)[28];
2. Indirect baroreflex-mediated HRV measured by low-frequency HRV (LF-HRV)[28].

Diverse interpretations and use of these sub-measures among researchers have resulted in disparate findings and a discordant literature[27]. The interpretation of HRV measures are further complicated by various factors such as participants' stress levels and required executive processing during experiment tasks, as well as other factors that can influence autonomic pain reactions (e.g., ethnicity, sex, age, body mass index, breathing patterns, stimulation intensity, affective states)[19]. Exploratory and nonconfirmatory research may help create testable hypotheses for future research in order to elucidate this complex relationship between HRV and pain[29]. Synthesizing data in a meta-analysis may contribute to a more comprehensive understanding of the HRV-pain relationships via increased statistical power and enabling informative tests of potential moderator variables[30,31].

This planned systematic review and meta-analysis aims to contribute to a more thorough characterization of the relationship between HRV and pain, and thus help patients who cannot otherwise communicate that they are in pain. Additionally, a better understanding of the mechanisms underlying how the autonomic nervous system responds to acute and chronic pain are vital for the further development of strategies to improve pain treatments, such as breathing exercises[32,33] or transcutaneous nerve stimulation[34]. In this way, findings may contribute to both improving pain detection and pain treatment.

The objective of this study is to assess the utility of HRV as an indicator of acute and long-term pain. To adequately address this objective, we will explore the HRV-pain

relationship using two kinds of HRV measures: Vagally-mediated HRV; and indirect baroreflex-mediated HRV. We have formulated two research questions: 1) What is the relationship between HRV and experimentally induced pain? 2) How do chronic pain patients compare to healthy individuals with respect to HRV? We hypothesize that chronic pain patients will have lower vagally-mediated HRV compared to control groups.

To address these questions, we will gather data from multiple studies, and we will deploy recently developed methods for detecting and correcting for publication bias. Altogether, this meta-analysis may contribute to a more comprehensive understanding of the relationship between HRV and pain. This systematic review and meta-analysis are pre-registered on PROSPERO (ID: CRD42023416895). This protocol follows the Statement for Reporting Literature Searches in Systematic Reviews from Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA-P) guidelines[35] (Supplementary file 1).

Methods

Eligibility Criteria

Inclusion: A study must meet the following criteria: 1) The study collects HRV data (RSA, RMSSD, HF power, and/or LF-HRV) for patients with a pain-related condition, and/or persons with no pain-related condition; and 2) data is reported on either a) HRV in pain-inducing procedure, or b) comparison of HRV in patients with a pain-related condition versus healthy controls. This way, studies are grouped into two for the data syntheses: 1) experimentally induced pain (within-subject studies) and 2) difference in HRV between persons with a chronic pain condition and persons with no chronic pain condition (between-subject studies).

Exclusion: Studies that are not conducted in humans or not reported in either English, Norwegian, or German will be excluded during screening.

Sample Size

Sensitivity power analysis for meta-analysis can be done prospectively, requiring assumptions about parameters that are unknown. However, as with primary studies, it is useful to determine the sample size needed to have sufficient power to detect the effect size of interest[36,37]. With typical sample sizes of 30 participants per group, a hypothetical true effect size of Hedges' $g = 0.48$, an expected moderate degree of heterogeneity and 80% power, the target number of studies to be included is eight[36]. The effect size of 0.48 was selected as it represents the median effect size for 73 HRV case-control studies evaluating pain[38].

Search Procedure

We will follow the PRISMA-P statement[39]. The exact search string and search strategy is available as Supplementary file 2.

Papers will be searched in PubMed/Ovid Medline, Embase, and Cochrane Central Register of Controlled Trials using controlled vocabulary and text words expressing HRV and pain. No date or language restrictions will be made. Additional searches will be performed in Bielefeld Academic Search Engine, as this search engine can facilitate the search of various preprint servers.

Search terms were developed with the assistance of a local academic librarian at University of Oslo library and customized for each database.

Data extraction and management

Study selection

First review author (EK) will independently evaluate eligibility of each identified study and select studies that will be included in meta-analysis using the Covidence systematic review management tool[40]. Reasons for exclusion will be noted.

The screening will be performed in two stages.

(1) In the first stage, we will search titles, abstracts, and full-text articles using search terms that are developed for each database. Author EK will screen titles, abstracts, and full-text articles of identified studies to determine studies that meet inclusion criteria. Any duplicate copies of studies will be removed using Zotero's built-in duplication function (zotero.org).

(2) In the second stage, we will search through the reference lists within studies identified from first screening and articles that cited the identified studies using Google Scholar.

Data extraction

Data will be collected from each study to calculate the standardized mean difference between conditions, and correlation. This can include means, standard deviations, group sizes, test statistics, and *p*-values. If data for effect size calculation cannot be extracted from the article text directly, we will attempt extraction from figures using WebPlotDigitizer[41]. If this is not successful, we will contact authors requesting the data needed for effect size calculation.

The following data from each study will also be extracted:

(1) Publication year;

(2) Number of participants;

(3) Characteristics of participants: chronic pain condition vs no pain condition, mean age, sex, ethnicity, recruitment strategy/place;

(4) Main study focus;

(5) Information about pain studied (naturally occurring vs experimentally induced, acute vs. chronic pain, musculoskeletal vs. visceral pain);

(6) Information about experiment (if experimental): method of pain induction, pain assessment;

(7) Information on HRV measure derived (RSA, RMSSD, HF power, LF-HRV), duration of HRV-sampling (long (~24 h), short-term (~5 min), and ultra-short-term (<5 min)), and sensor type used (electrocardiogram: ECG, photoplethysmography: PPG);

(7) Outcome measurement: different methods of measuring heart rate variability (e.g., resting vs. during a stress test) and pain intensity (e.g., self-report vs. physiological measures);

(8) Peer-review status (e.g., peer-reviewed journal article or preprint);

(9) Information on pre-registration or lack thereof.

A second author (KA) will also extract data from a random 50% sample of the eligible studies. If there is any disagreement with data extracted, this will be resolved by a discussion mediated by a third author (DSQ) and then data from the remainder 50% of the studies will also be extracted.

Handling missing data

We will record studies with missing data and contact the authors via email for data request either in the form of raw data or computed effect sizes (e.g., Cohen's *d* or Hedges' *g*).

Data quality, small study bias and publication bias

Risk of bias in individual studies

We will assess risk of bias for each randomized intervention study using RoB 2: a revised tool for assessing risk of bias in randomized trials[42]. Two reviewers (EK and KA) will independently take part in the assessment and discuss to reach consensus upon any disagreement. The statistical power of studies included in the meta-analysis for a range of effect sizes will also be calculated and visualized using the metameta R package[43].

For correlational studies, we will provide descriptive information on the criteria's of RoB 2 that are also relevant for correlational studies.

Reporting bias

We will formally assess the presence of small study bias using a contour-enhanced funnel plot visualization and Egger's regression test[44]. Additionally, we will perform sensitivity analysis using the R package PublicationBias to evaluate the potential severity of publication bias in our study[44].

Statistical analyses

All statistical analyses will be implemented using R[45] and R studio[45]. The R package metafor[46] will be used for meta-analysis and the TOSTER R package[47] will be used for equivalence testing of the experimental studies and the summary effect size.

Summary effect size

Eligible studies for our analyses are expected to have varying designs. Therefore, we will use different methods to calculate effect sizes for data synthesis in experimental- and correlational studies.

For experimental studies, we will calculate standardized mean difference, as Hedges' *g*. Calculation of standardized mean change in experimental studies requires the between-measurements correlation[48]. We will calculate between-measure correlation for studies where raw data is available, and use these as proxy for the studies not providing raw data. Next, we will compute the summary effect size and 95% confidence interval using a random-effects model, which accounts for heterogeneity among individual studies in terms of study population and design. To address the potential issue of multiple reports from the same study participants using different measures or tasks, we will employ a multi-level meta-analysis approach.

For correlational studies, we will calculate Fisher's z transformed correlations. After analysis, the correlations will be back-transformed to Pearson's *r* for easier interpretation.

Sample heterogeneity

Under a random-effects model, any observed difference between individual studies may consist of both random error within studies and heterogeneity in effect sizes between studies. This between-study heterogeneity will be tested and reported using Cochran's *Q* and I^2-statistic.

Moderator variables

Variables such as sensor type and duration of HRV-sampling introduce systematic variations can influence the observed associations between HRV and pain. We will include 1) sensor type: electroencephalography (EEG) vs. photoplethysmography (PPG); and 2)

duration of HRV sampling (short-term <10 min vs. long-term); as moderator variables. This allows the assessment of their impact on the relationship, and thus may improve our understanding of the HRV-pain relationship.

Equivalence testing

Summary effect sizes that are not statistically significant do not necessarily reflect the absence of an effect, as this may indicate insufficient statistical power to detect small effect sizes[47]. We will therefore calculate statistical equivalence, which evaluates the evidence for rejecting an effect size that is considered meaningful, otherwise known as the smallest effect size of interest[47]. The equivalence bound is set to ±0.12 as this corresponds to the 25th percentile of effect sizes across 73 studies on HRV and pain[38] (i.e., small effect size).

Discussion

In this planned systematic review and meta-analysis, we aim to systematically investigate the relationship between HRV and pain. By synthesizing available data from multiple studies, we hope to provide a more comprehensive understanding of this relationship.

The results of this meta-analysis may uncover possibilities regarding the real-time use of HRV. Specifically, if the findings indicate a strong relationship between HRV and pain, it could support the potential utilization of HRV to aid patients who cannot otherwise communicate that they are in pain (e.g., aphasic stroke patients, persons with profound autism, cerebral palsy, or profound intellectual disability). Thus, insights from this systematic review and meta-analysis may inform the development of improved detection strategies, tailored pain management strategies, and ultimately lead to improved patient quality of life.

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Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this preprint article.

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Author contributions

The study idea was conceived by EK and DSQ. All authors (EK, KA, HBJ, HK, AS, BG, ED, AS, CT, OA, LW, BH, TN, DSQ) have contributed substantially to the present work. The first draft was written by EK. All authors (EK, KA, HBJ, HK, AS, BG, ED, AS, CT, OA, LW, BH, TN, DSQ) have added to the previous versions of the manuscript. All authors (EK, KA, HBJ, HK, AS, BG, ED, AS, CT, OA, LW, BH, TN, DSQ) have read and approved the final manuscript. The guarantors of the review are EK and DSQ.

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